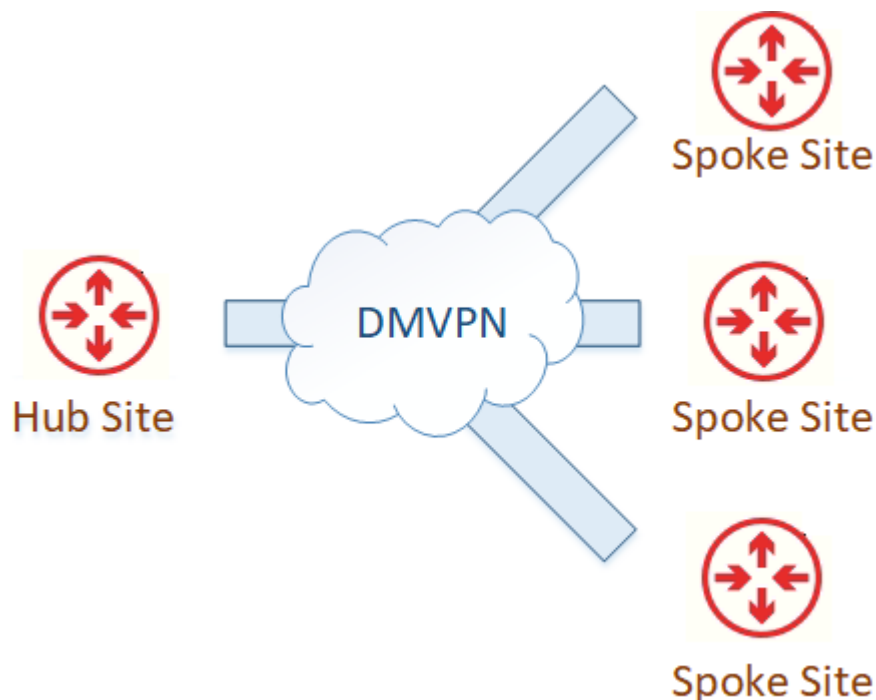


DMVPN:

- o DMVPN is a term, which stands for **Dynamic Multipoint Virtual Private Network**.
- o DMVPN is Cisco solution to build easy, dynamic and scalable IPsec and GRE VPNs.
- o DMVPN is used to connect multiple sites in Partial or Full Mesh network topology.
- o DMVPN is Cisco tunnel technology with Hub-and-Spoke and Spoke to Spoke tunnels.
- o Hub-to-Spoke tunnels are always up, Spoke-to-Spoke are "On Demand" dynamically.
- o Cisco DMVPN's are a highly scalable, and easy to manage tunnel solution and method.
- o Similar in many ways to traditional tunnels like IPsec site-to-site VPN, or a GRE tunnel.
- o DMVPN does not have the static nature of traditional VPNs like IPsec Site-to-Site VPN.
- o Traditional tunnels are point-to-point, two routers connect together with virtual tunnel.
- o Dynamic Multipoint Virtual Private Network allows to configure single tunnel at Hub Site.
- o Each spoke site can join DMVPN network, with minimal effort, & dynamic IP addresses.
- o DMVPN keeping costs low, minimizing configuration complexity and increasing flexibility.
- o DMVPN, one central Router, usually placed at the head office, undertakes the role of Hub.
- o Other branch Routers are Spokes connect to Hub Router to access company's resources.
- o Hub Router undertakes the role of the server while the spoke routers act as the clients.
- o Hub maintains special NHRP database with public IP Addresses of all configured spokes.
- o The hub must have a known (Static) IP, while the spokes may use dynamic addressing.
- o To secure the traffic passed over the Internet, IPSEC is used to encrypt the packets.



DMVPN Deployment:

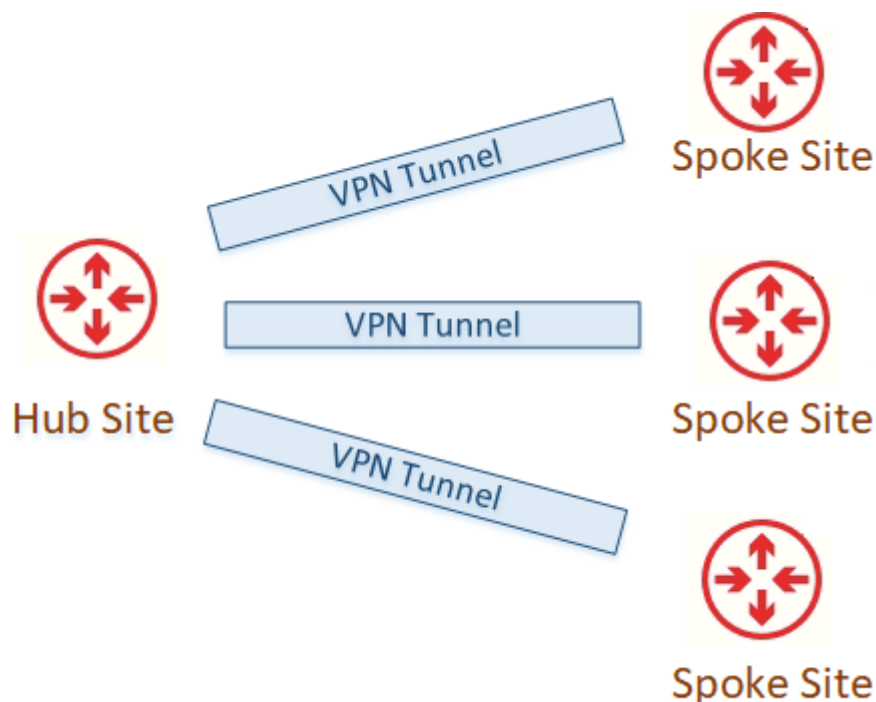
- o Dynamic Multipoint Virtual Private Network can be deployed in three different way.
- o Dynamic Multipoint Virtual Private Network can be deployed in phase 1, phase 2 or phase 3.
- o Phase 1: Hub and Spoke mGRE at the Hub and Point-to-Point p2p GRE at Branch Routers.
- o Phase 2: Hub and Spoke with Spoke-to-spoke tunnels (mGRE at the Hub & Branch Routers).
- o Phase 3: Same as Phase 2 but now Spoke Routers can also respond to NHRP Requests.
- o Phase 2 allow spokes to build a spoke-to-spoke tunnel on demand with some restrictions.
- o DMVPN Phase 2 supports spoke to spoke but requires a certain routing design to apply.
- o Phase3 allow spokes to build spoke-to-spoke tunnel & overcomes the phase 2 restriction.
- o Phase 3 is latest & flexible design supporting both hub-to-spoke & spoke-to-spoke tunnels.
- o DMVPN using three different technologies- mGRE, Next-Hop Resolution Protocol & IPSec.

DMVPN Hub-to-Spoke:

- o DMVPN Hub-to-Spoke used to HUB site (Data Center) to Remote site communication.
- o Phase 1 Hub and Spoke - mGRE at the Hub and point to point GRE at the Branch routers.

DMVPN Spoke-to-Spoke:

- o DMVPN Spoke-to-Spoke used to perform Remote site to Remote Site direct communication
- o Phase 2 Hub and Spoke with Spoke-to-spoke tunnels (mGRE at the Hub and Branch routers).
- o Phase 3 Same as Phase 2 but now Spoke routers can also respond to NHRP Requests.



DMVPN Terminology:

NBMA Address:

- o NBMA stands for Non-Broadcast Multiple Access Address.
- o NBMA IP Address, typically a public IP address on Internet facing interface.
- o NBMA IP Address is uuter IP header address also called Underlay or transport.
- o Public IP Address, Infrastructure IP address, Outside IP or Service Provider Address.

Next Hop Server (NHS):

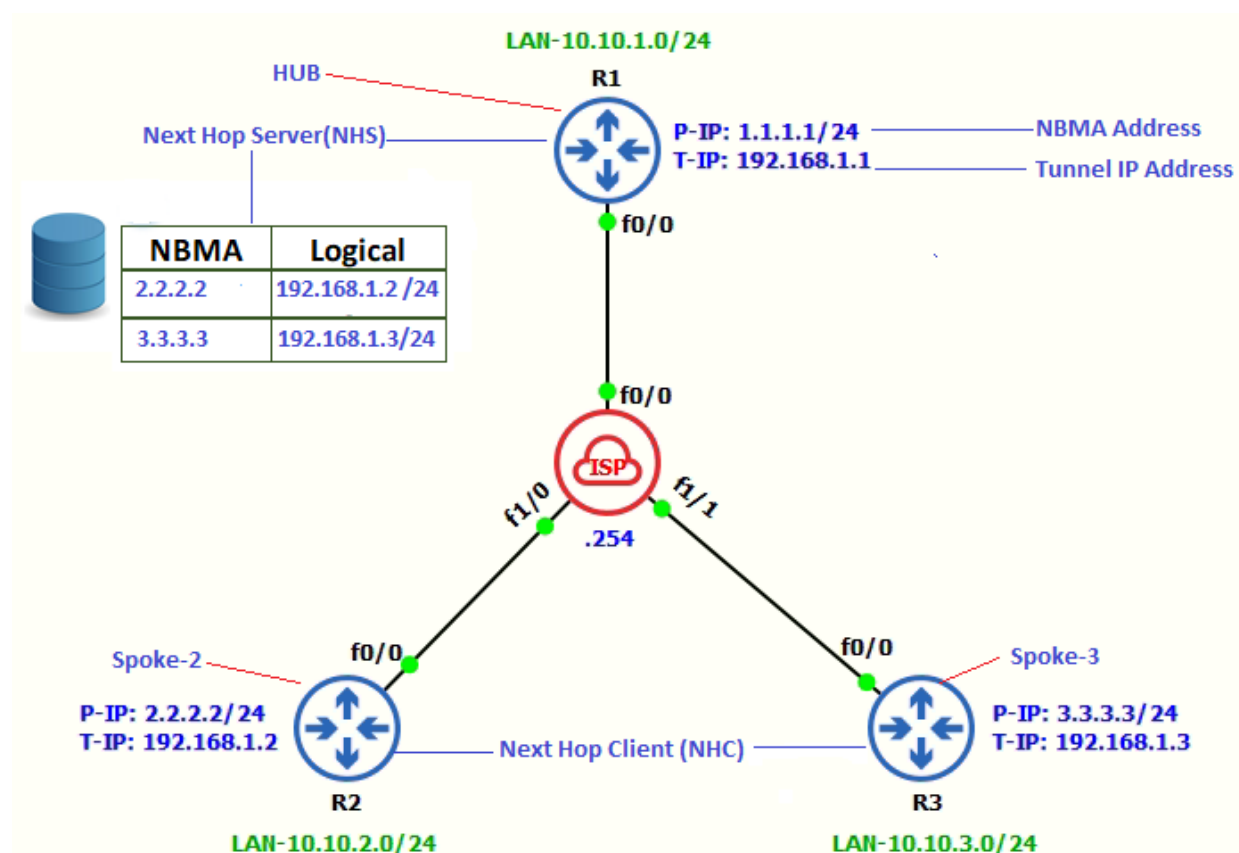
- o Dynamic Multipoint Virtual Private Network Hub Router(s) or Router on the Hub site.
- o Stores NBMA/protocol address mapping information received via NHRP registration.

Next Hop Client (NHC):

- o Dynamic Multipoint Virtual Private Network Spoke Router(s) or Router on Spoke site.
- o Registers its NBMA/protocol address mapping to NHS with NHRP registration messages.

Tunnel IP Address:

- o Generic Routing Encapsulation (GRE) Tunnel interface IP address called Overlay.
- o Logical Tunnel interface address used in overlay network also called Protocol address.
- o Private IP Addresses are also called Enterprise Addressing space, Inside IP address.



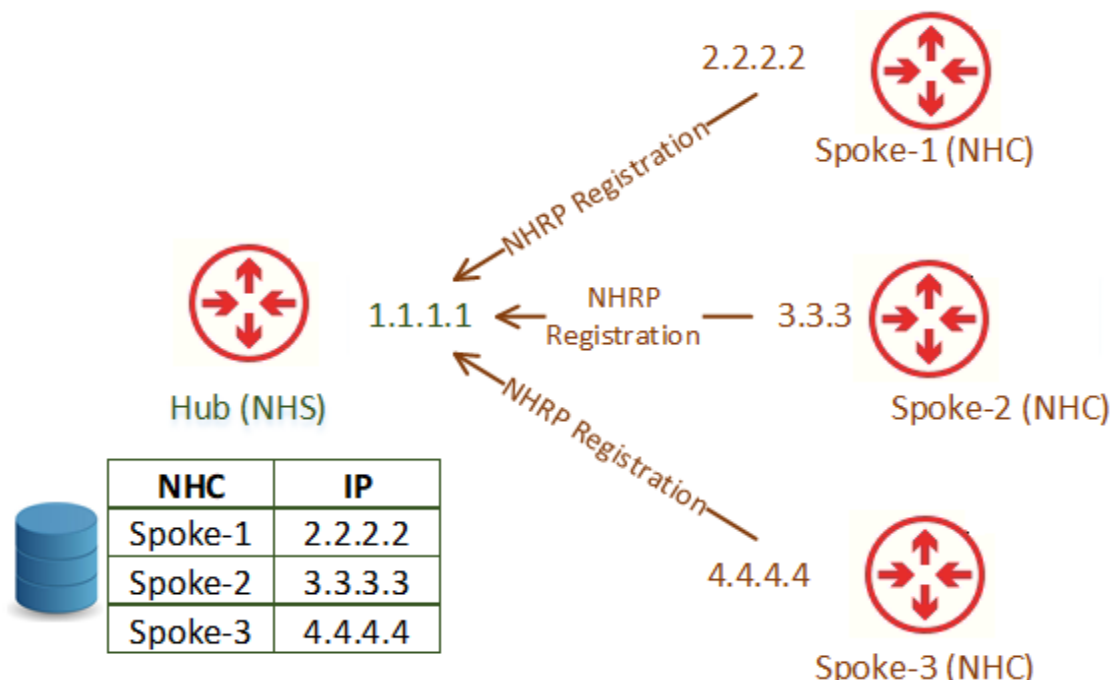
DMVPN Components:

GRE, mGRE (Multipoint GRE):

- o DMVPN uses mGRE tunneling mode for building dynamic Point-to-multipoint tunnels.
- o mGRE serves as one-to-many interface for the creation of multiple hub-and-spoke tunnels.
- o The tunnel destination for Multipoint GRE (mGRE) tunnel does not have to be configured.
- o Multipoint GRE (mGRE) tunnel is effectively NBMA medium like Frame-Relay.
- o All the Hub routers are configured with Multipoint GRE (mGRE) tunnel interface.
- o If Hub-and-Spoke model is to deploy, Branch routers are configured with p2p GRE tunnel.
- o If Spoke-to-Spoke model is to deploy, Branch routers must be configured with mGRE tunnel.

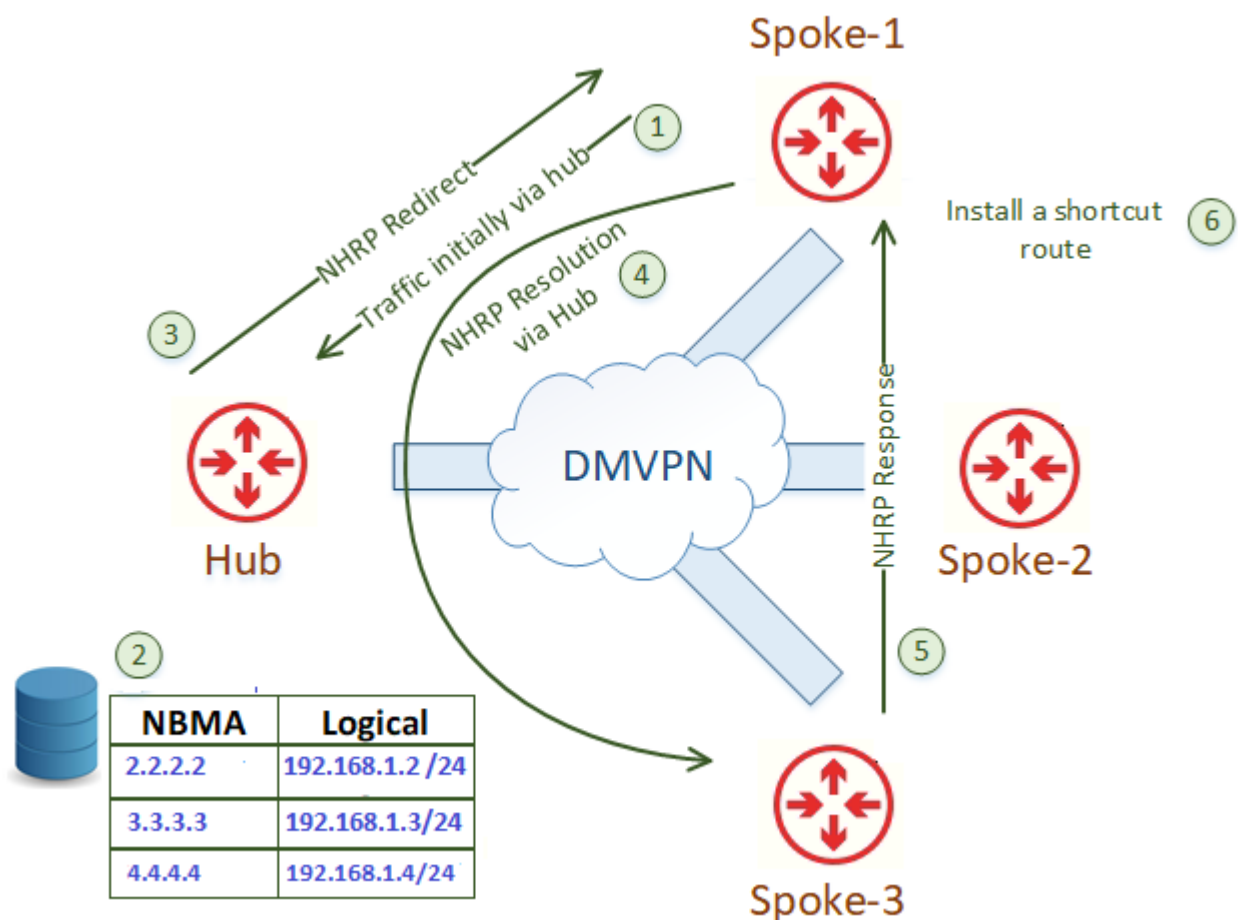
NHRP (Next Hop Resolution Protocol):

- o NHRP is a Layer-2 address resolution protocol like ARP and Frame-relay Inverse-ARP.
- o NHRP is used by Branch router connected to NBMA network to determine IP of Next-Hop.
- o All Branch routers register their NBMA IP mapped to logical IP addresses with NHS.
- o The IP address of the Next-Hop Server (NHS) is pre-configured on the Branch routers.
- o The NHS stores all registered mappings and replies to NHRP Requests from NHCs.
- o NHRP allows mGRE tunnel endpoints to discover each other's physical IP address.
- o NHRP facilitates in building Multipoint Generic Routing Encapsulation (mGRE) tunnels.
- o NHRP is not a new protocol being used with NBMA networks like Frame-Relay and ATM.
- o NHRP uses a client/server model hub router is the NHS, spoke routers are the NHC.
- o All NHRP messages are encapsulated in GRE and sent to the logical tunnel IP Address.
- o NHRP provides address resolution in NBMA networks, similar to ARP address resolution.
- o NHRP maps a logical VPN IP to a public NBMA IP without the need for broadcast.
- o NHRP, at its core protocol, which resolves private addresses to a public address.



DMVPN Works:

- o Initially each spoke establish a permanent GRE + IPsec tunnel to the hub.
- o The hub address should be static and known by all the spokes routers.
- o Each spoke registers its NBMA IP Address as a client to the NHRP server on the hub.
- o NHRP server maintains an NHRP database of public interface IP address for each spoke.
- o When a spoke requires that packets be sent to a destination subnet on another spoke.
- o It sends query to NHRP server to get information of NBMA address of the other spoke.
- o When get information of NBMA Address of other spoke so that it can build direct tunnel.
- o NHRP server looks up NHRP database for corresponding destination spoke and replies.
- o After originating spoke learns NBMA IP of target spoke, it initiates dynamic IPsec tunnel.
- o With integration of mGRE interface, NHRP & IPsec, direct dynamic tunnel is established.
- o Adding a new spoke router to the DMVPN requires no configuration on the hub router.
- o Spoke is configured with hub information & dynamically registers with the hub router.

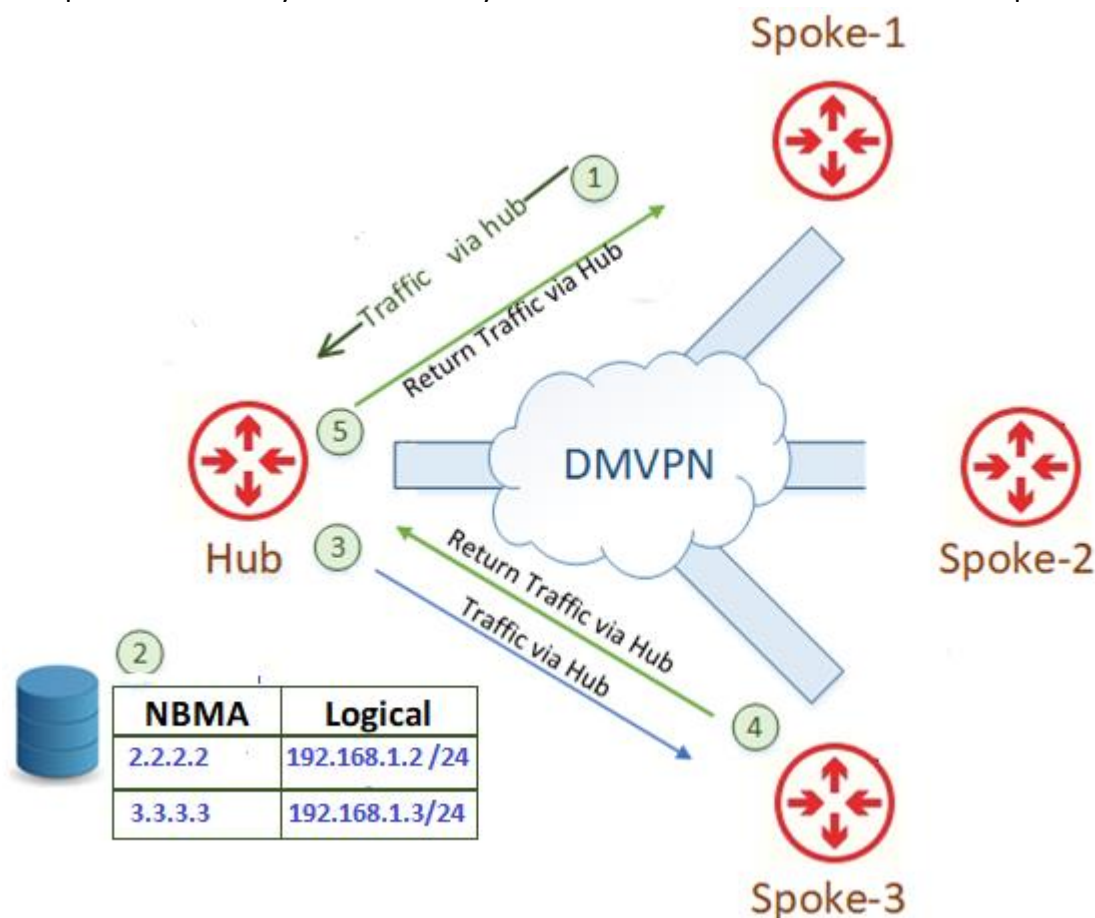


DMVPN Network Designs:

- o DMVPN can be deployed in three different ways, these are called **phases**.
- o These are not phases like in IPSec, they are also not stages that the tunnels go through.
- o These are like version numbers, for example, Phase 2 has different features to phase 1.

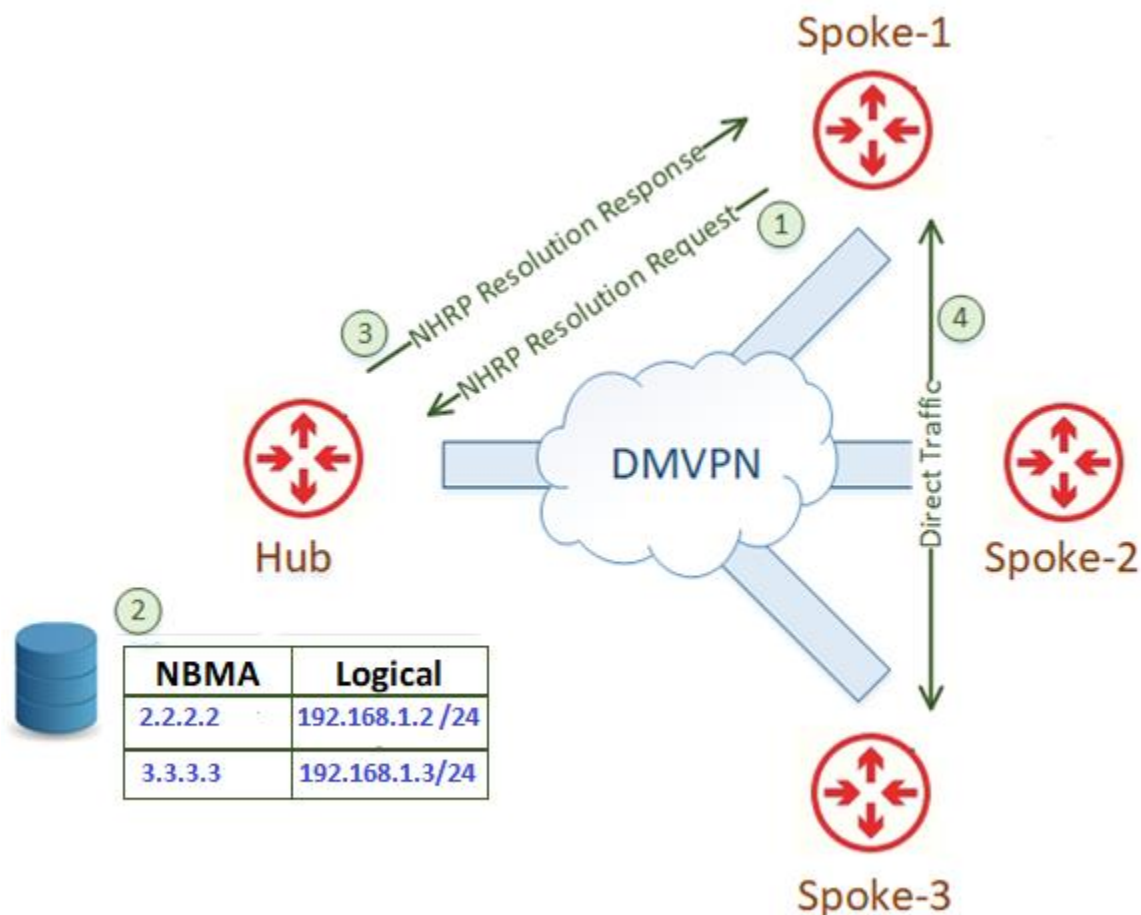
Phase 1 – Hub to Spoke Design:

- o Phase 1 was original implementation of DMVPN, based entirely on hub & spoke model.
- o Spokes use NHRP & register with hub, hub router builds mGRE tunnel to connect all spokes.
- o In DMVPN each spoke builds a regular GRE tunnel, and will only connect to the hub router.
- o mGRE tunnel is configured at Hub and GRE tunnel is configured on the spoke's routers.
- o Phase 1 DMVPN allows spoke routers to dynamically register with hub routers.
- o Hub to Spoke design Multicast or Unicast traffic flow between Hub & Spokes only.
- o In DMVPN Phase 1 Hub to Spoke Design No Spoke-to-Spoke traffic flow is allowed.
- o In DMVPN Phase 1 Hub to Spoke, Spoke-to-Spoke traffic traverse via the Hub router.
- o In Phase 1 there is no way for spokes to directly communicate with each other.
- o If spoke-1 wants to communicate with Spoke-3, traffic needs to flow via the hub.
- o traffic arrives at hub, needs to be decapsulated, then be encapsulated again & sent.
- o Spoke routers only need summary or default route to hub to reach other spoke routers.



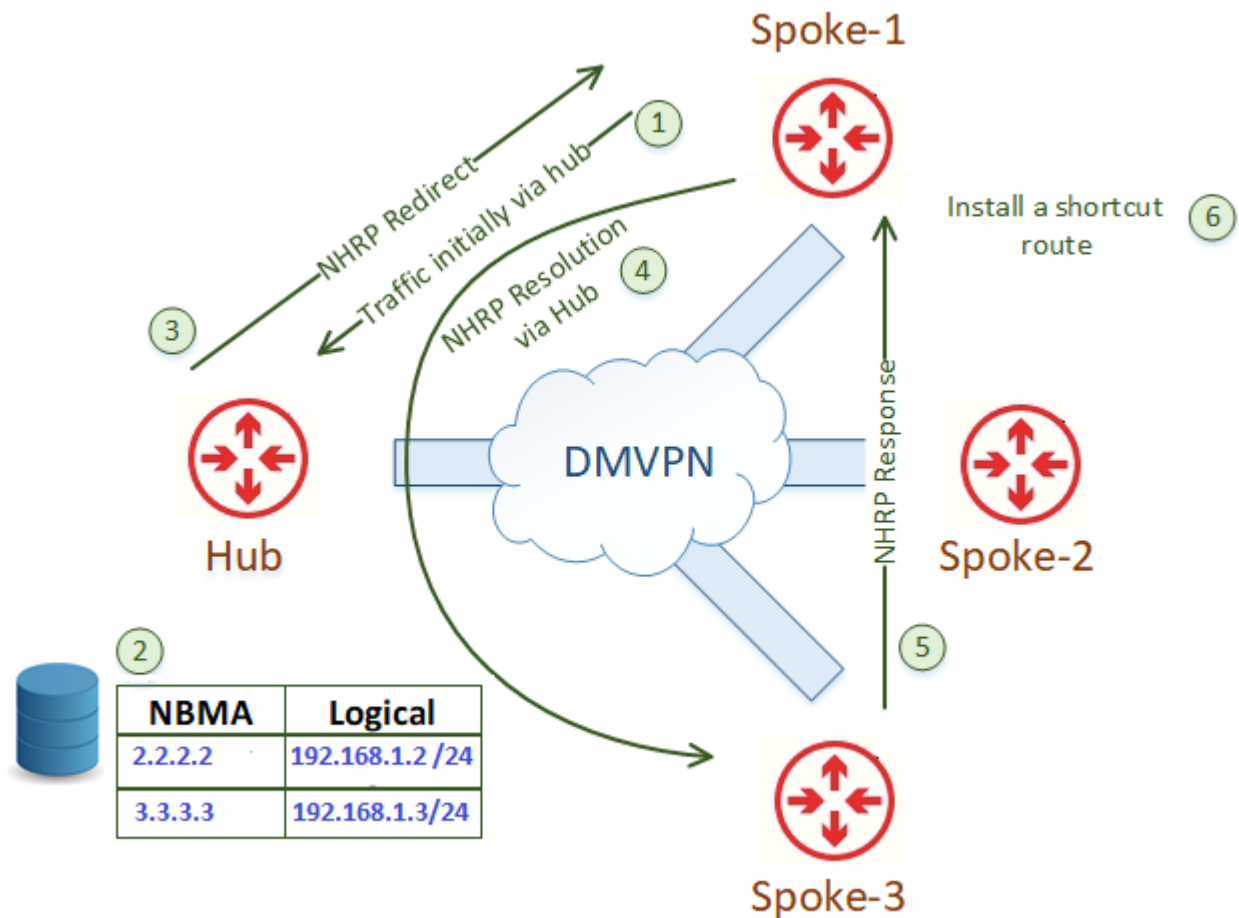
Phase 2 – Spoke to Spoke Design:

- o Biggest limitation in Phase 1 was the lack of direct communication between spokes.
- o Phase 2 is an improvement on Phase 1, It retains dynamic spoke registration.
- o Add mGRE on spoke side, this means enables direct communication between spokes.
- o Phase-2 Spoke routers can now communicate directly without traffic traversing hub.
- o In DMVPN Phase 2, Hub and all Spokes will be configured with mGRE tunnel.
- o In DMVPN Phase 2, Spoke-to-Spoke traffic traverse via direct dynamic VPN tunnel.
- o In DMVPN Phase 2, Spoke starts by sending an NHRP Resolution Request to the hub.
- o The hub looks into its cache of NBMA address mappings and finds the correct entry.
- o Router caches info request & then sends NHRP Resolution Response back to original spoke.
- o The response contains the NBMA address details of the destination spoke router.
- o The spoke router now knows the NBMA address details of the remote spoke router.
- o In DMVPN the hub forwards the resolution request to the remote spoke router.
- o In DMVPN Phase 2, Route Summarization not feasible at Hub Router.

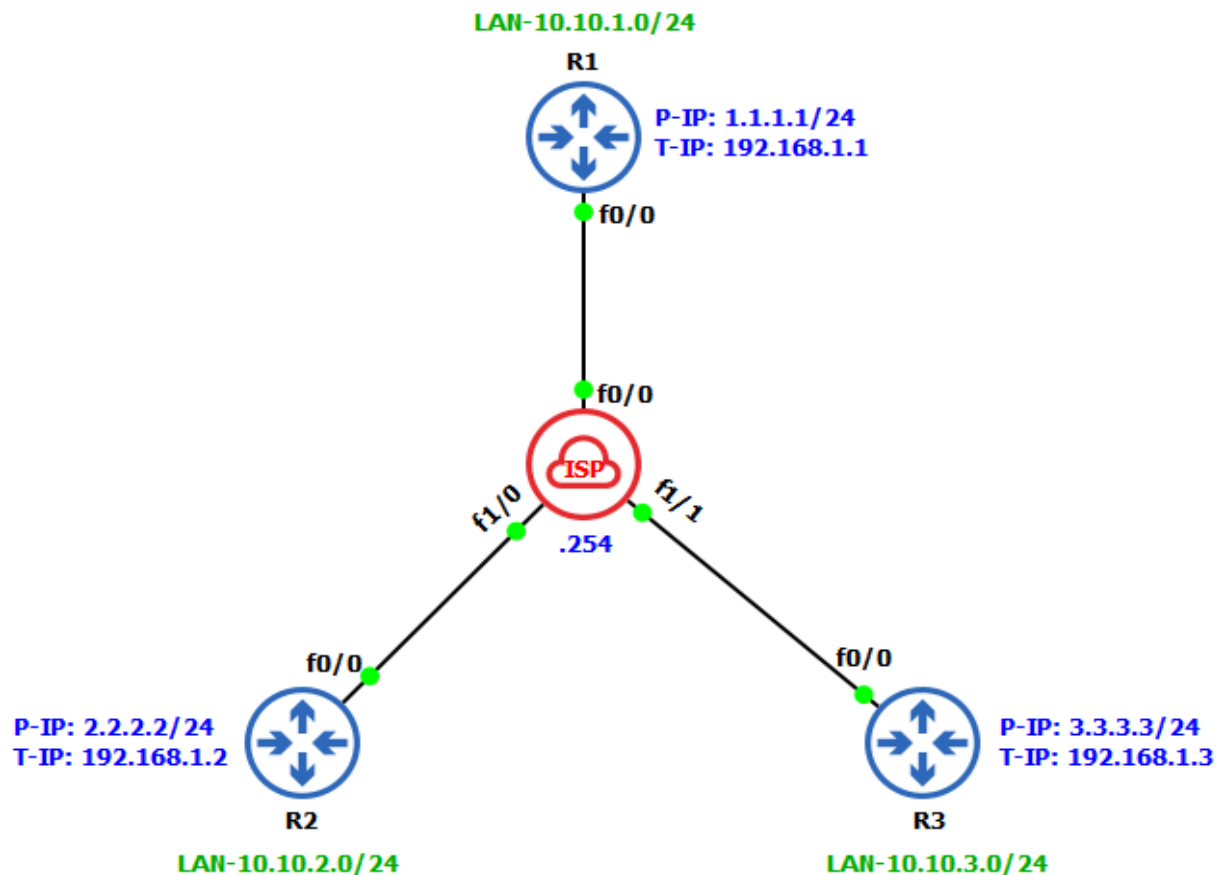


Phase 3 – Spoke to Spoke Design:

- o DMVPN Phase 3 is the final and most scalable phase in Dynamic Multipoint VPN.
- o On spokes only need to enable NHRP shortcuts for Dynamic Multipoint VPN phase 3.
- o Phase-3 adds efficient route summarization & allowing Spoke-to-Spoke communication.
- o The spoke will no longer start with an NHRP Resolution Request but sending traffic directly.



DMVPN Phase 1 Configuration:



R1 Router Basic Configuration

```
R1(config)#interface f0/0
R1(config-if)#ip add 1.1.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config)#interface loopback1
R1(config-if)#ip add 10.10.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config)#ip route 0.0.0.0 0.0.0.0 1.1.1.254
```

R2 Router Basic Configuration

```
R2(config)#interface f0/0
R2(config-if)#ip add 2.2.2.2 255.255.255.0
R2(config-if)#no shutdown
R2(config)#interface loopback1
R2(config-if)#ip add 10.10.2.1 255.255.255.0
R2(config-if)#no shutdown
R2(config)#ip route 0.0.0.0 0.0.0.0 2.2.2.254
```

R3 Router Basic Configuration

```
R3(config)#interface f0/0
R3(config-if)#ip add 3.3.3.3 255.255.255.0
R3(config-if)#no shutdown
R3(config)#interface loopback1
R3(config-if)#ip add 10.10.3.1 255.255.255.0
R3(config-if)#no shutdown
R3(config)#ip route 0.0.0.0 0.0.0.0 3.3.3.254
```

ISP Router Basic Configuration

```
ISP(config)#interface f0/0
ISP(config-if)#ip add 1.1.1.254 255.255.255.0
ISP(config-if)#no shutdown
ISP(config)#interface f1/0
ISP(config-if)#ip add 2.2.2.254 255.255.255.0
ISP(config-if)#no shutdown
ISP(config)#interface f1/1
ISP(config-if)#ip add 3.3.3.254 255.255.255.0
ISP(config-if)#no shutdown
```

R1 Hub Router Phase 1 Configuration

```
R1(config)#interface Tunnel 0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no ip redirects
R1(config-if)#tunnel mode gre multipoint
R1(config-if)#tunnel source f0/0
R1(config-if)#ip nhrp authentication vpn
R1(config-if)#ip nhrp map multicast dynamic
R1(config-if)#ip nhrp network-id 1
```

```
R1#show dmvpn
```

```
R1#show ip nhrp
```

R2 Spoke Router Phase 1 Configuration

```
R2(config)#interface tunnel 0
R2(config-if)#ip address 192.168.1.2 255.255.255.0
R2(config-if)#ip nhrp authentication vpn
R2(config-if)#ip nhrp map 192.168.1.1 1.1.1.1
R2(config-if)#ip nhrp map multicast 1.1.1.1
R2(config-if)#ip nhrp network-id 1
R2(config-if)#ip nhrp nhs 192.168.1.1
R2(config-if)#tunnel source f0/0
R2(config-if)#tunnel destination 1.1.1.1
```

```
R2#show dmvpn
```

```
R1#show ip nhrp
```

R3 Spoke Router Phase 1 Configuration

```
R3(config)#interface tunnel 0
R3(config-if)#ip address 192.168.1.3 255.255.255.0
R3(config-if)#ip nhrp authentication vpn
R3(config-if)#ip nhrp map 192.168.1.1 1.1.1.1
R3(config-if)#ip nhrp map multicast 1.1.1.1
R3(config-if)#ip nhrp network-id 1
R3(config-if)#ip nhrp nhs 192.168.1.1
R3(config-if)#tunnel source f0/0
R3(config-if)#tunnel destination 1.1.1.1
```

```
R3#show dmvpn
```

```
R3#show ip nhrp
```

```
R3#traceroute 192.168.1.2
```

No IP Redirects:

- o Disables ICMP Redirects on this interface.
- o If the next-hop address isn't set as hub cause ICMP redirect message to be sent.
- o Disables ICMP Redirects prevent excessive redirect traffic.

Tunnel Mode GRE Multipoint:

- o This command specifies the interface as a multipoint GRE interface.
- o This command should always be configured on the HUB for phase 1.
- o by default, tunnel mode point-to-point GRE, require a multipoint interface on hub.

Tunnel Source F0/0:

- o This command specifies the source of the tunnel interface.
- o this should be set to the public interface which is connected to your ISP.
- o the tunnel destinations will be dynamic, but we still have to configure the source.
- o The address of this interface will be advertised in the registration message.
- o This tunnel source interface should be reachable via the spokes.

IP NHRP Authentication:

- o This command requires spokes to authenticate with hub before registration.
- o This command requires to authenticate the NHRP traffic its optional.
- o This command require for spokes to resolution requests can be made.

IP NHRP Map Multicast Dynamic:

- o This command tells the hub router where to forward multicast packets to.
- o the IP addresses of spoke routers are unknown, use dynamic to automatically add.
- o Their IP addresses to the multicast destination list when spokes register themselves.
- o This command allows to send all multicast traffic to all dynamically learned spokes.

IP NHRP Network-ID:

- o NHRP Network-ID command specifies the ID of the DMVPN cloud.
- o It's required to allow a router to distinguish between each DMVPN network.
- o In multiple DMVPN networks, need network ID to differentiate between two networks.
- o Command identifies the logical network, this must be same on all NHRP routers.

IP NHRP Map:

- o Use on spoke to create static mapping for the hub's tunnel address NBMA address.
- o NHRP details will be stored in the NHRP cache of the spoke router.

IP NHRP Map Multicast:

- o Which destinations should receive multicast traffic through tunnel interface.
- o IP nhrp map multicast command have to enter the NBMA address here.
- o Need this command since routing protocols RIP, EIGRP & OSPF require multicast.

IP NHRP NHS:

- o This command is where we specify the NHRP server, our hub router.
- o This command specifies who the next hop server is on the network.
- o This is VPN address and therefore a static mapping for this will be needed.

Tunnel Destination:

- o Using point-to-point GRE need to specify destination for tunnel, which is hub router.

Routing on DMVPN Phase 1:

- o For EIGRP, need to disable split-horizon on Hub mGRE tunnel interface.
- o Disable split-horizon on Hub mGRE to ensure optimal forwarding.

No IP Split-Horizon EIGRP 1:

- o Disables split-horizon on hub so routes from one spoke can be sent to another.
- o EIGRP is loop-free due to the feasibility condition this will not cause a loop.

ip Summary-Address EIGRP 1 0.0.0.0 0.0.0.0:

- o Send out a default summary route to spokes.
- o Only hub & spoke traffic flows are allowed default routing can be done.
- o Via the hub to reduce the routing table information on spokes.

R1 Hub Router EIGRP Configuration

```
R1(config)#interface Tunnel 0
R1(config-if)# no ip split-horizon eigrp 1
R1(config-if)# ip summary-address eigrp 1 0.0.0.0 0.0.0.0
```

EIGRP Configuration on Hub Router

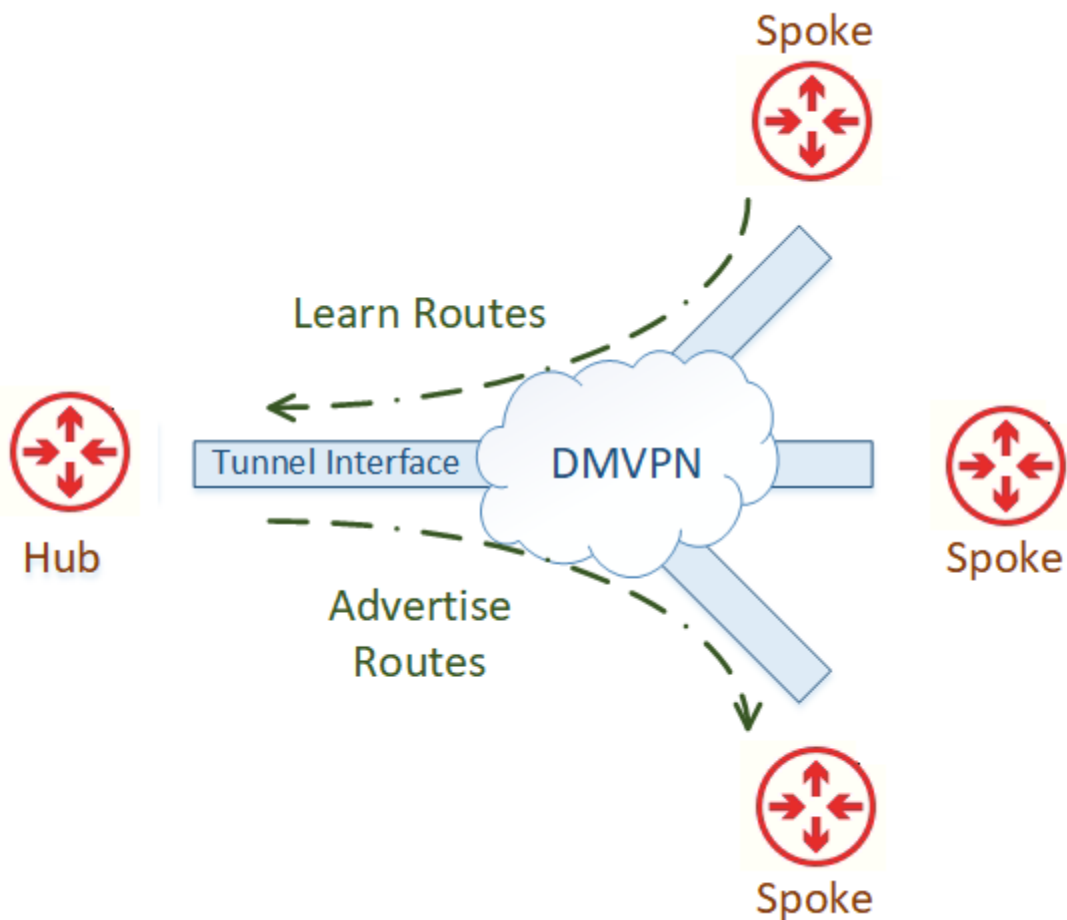
```
R1(config)#router eigrp 1  
R1(config-router)#network 10.10.1.0 0.0.0.255  
R1(config-router)#network 192.168.1.0  
R1(config-router)#no auto-summary
```

EIGRP Configuration on Spoke Router

```
R2(config)#router eigrp 1  
R2(config-router)#network 10.10.2.0 0.0.0.255  
R2(config-router)#network 192.168.1.0  
R2(config-router)#no auto-summary
```

EIGRP Configuration on Spoke Router

```
R3(config)#router eigrp 1  
R3(config-router)#network 10.10.3.0 0.0.0.255  
R3(config-router)#network 192.168.1.0  
R3(config-router)#no auto-summary
```



DMVPN Phase 2 Configuration:

- o The configuration of DMVPN phase 1 and 2 is similar except for two key items.
- o Spoke routers use multipoint GRE interfaces instead of point-to-point GRE interfaces.
- o Also, no need to configure a manual destination anymore on the spoke routers.
- o Removed tunnel destination command & added tunnel mode command GRE multipoint.
- o These two changes make the difference between running DMVPN phase 1 or phase 2.

R1 Hub Router Phase 2 Configuration
R1(config)#interface Tunnel 0 R1(config-if)#ip address 192.168.1.1 255.255.255.0 R1(config-if)#no ip redirects R1(config-if)#tunnel mode gre multipoint R1(config-if)#tunnel source f0/0 R1(config-if)#ip nhrp authentication vpn R1(config-if)#ip nhrp map multicast dynamic R1(config-if)#ip nhrp network-id 1
R1#show dmvpn
R1#show ip nhrp
R2 Spoke Router Phase 2 Configuration
R2(config)#interface tunnel 0 R2(config-if)#ip address 192.168.1.2 255.255.255.0 R2(config-if)#ip nhrp authentication vpn R2(config-if)#ip nhrp map 192.168.1.1 1.1.1.1 R2(config-if)#ip nhrp map multicast 1.1.1.1 R2(config-if)#ip nhrp network-id 1 R2(config-if)#ip nhrp nhs 192.168.1.1 R2(config-if)#tunnel source f0/0 R2(config-if)#no tunnel destination 1.1.1.1 R2(config-if)#tunnel mode gre multipoint
R3 Spoke Router Phase 2 Configuration
R3(config)#interface tunnel 0 R3(config-if)#ip address 192.168.1.3 255.255.255.0 R3(config-if)#ip nhrp authentication vpn R3(config-if)#ip nhrp map 192.168.1.1 1.1.1.1 R3(config-if)#ip nhrp map multicast 1.1.1.1 R3(config-if)#ip nhrp network-id 1 R3(config-if)#ip nhrp nhs 192.168.1.1 R3(config-if)#tunnel source f0/0 R3(config-if)#no tunnel destination 1.1.1.1 R3(config-if)#tunnel mode gre multipoint
R3#traceroute 192.168.1.2
R3#traceroute 192.168.1.2

Routing on Phase 2:

- o For EIGRP, need to disable split-horizon on Hub mGRE tunnel interface.
- o Disable Split-horizon on Hub tunnel Interface to ensure optimal forwarding.
- o Also need to disable the default EIGRP behavior to overwrite the next-hop as itself.

R1 Hub Router EIGRP Configuration

```
R1(config)#interface Tunnel 0
R1(config-if)# no ip split-horizon eigrp 1
R1(config-if)# no ip next-hop-self eigrp 1
```

EIGRP Configuration on Hub Router

```
R1(config)#router eigrp 1
R1(config-router)#network 10.10.1.0 0.0.0.255
R1(config-router)#network 192.168.1.0
R1(config-router)#no auto-summary
```

EIGRP Configuration on Spoke Router

```
R2(config)#router eigrp 1
R2(config-router)#network 10.10.2.0 0.0.0.255
R2(config-router)#network 192.168.1.0
R2(config-router)#no auto-summary
```

EIGRP Configuration on Spoke Router

```
R3(config)#router eigrp 1
R3(config-router)#network 10.10.3.0 0.0.0.255
R3(config-router)#network 192.168.1.0
R3(config-router)#no auto-summary
```

```
R1# show ip route
```

```
R1# show ip route eigrp
```

```
R2# show ip route
```

```
R2# show ip route eigrp
```

```
R3# show ip route
```

```
R3# show ip route eigrp
```

```
R3# ping 10.10.2.1 source loopback 1
```

```
R2# ping 10.10.3.1 source loopback 1
```

```
R3# traceroute 10.10.2.1 source loopback 1
```

```
R3# traceroute 10.10.3.1 source loopback 1
```


DMVPN Phase 3 Configuration:

- o To change from Dynamic Multi VPN phase 2 to phase 3, only need two commands.
- o There are two commands NHRP redirects on the hub, and shortcut routes on the spokes.
- o NHRP redirect enables the hub to inform a spoke of a better path if one exists.
- o NHRP shortcut command allows spoke to accept redirect message install shortcut route.

R1 Hub Router Phase 3 Configuration
R1(config)#interface Tunnel 0 R1(config-if)#ip address 192.168.1.1 255.255.255.0 R1(config-if)#no ip redirects R1(config-if)#tunnel mode gre multipoint R1(config-if)#tunnel source f0/0 R1(config-if)#ip nhrp authentication vpn R1(config-if)#ip nhrp map multicast dynamic R1(config-if)#ip nhrp network-id 1 R1(config-if)#ip nhrp redirect
R1#show dmvpn
R1#show ip nhrp
R2 Spoke Router Phase 3 Configuration
R2(config)#interface tunnel 0 R2(config-if)#ip address 192.168.1.2 255.255.255.0 R2(config-if)#ip nhrp authentication vpn R2(config-if)#ip nhrp map 192.168.1.1 1.1.1.1 R2(config-if)#ip nhrp map multicast 1.1.1.1 R2(config-if)#ip nhrp network-id 1 R2(config-if)#ip nhrp nhs 192.168.1.1 R2(config-if)#tunnel source f0/0 R2(config-if)#tunnel mode gre multipoint R2(config-if)# ip nhrp shortcut
R3 Spoke Router Phase 3 Configuration
R3(config)#interface tunnel 0 R3(config-if)#ip address 192.168.1.3 255.255.255.0 R3(config-if)#ip nhrp authentication vpn R3(config-if)#ip nhrp map 192.168.1.1 1.1.1.1 R3(config-if)#ip nhrp map multicast 1.1.1.1 R3(config-if)#ip nhrp network-id 1 R3(config-if)#ip nhrp nhs 192.168.1.1 R3(config-if)#tunnel source f0/0 R3(config-if)#tunnel mode gre multipoint R2(config-if)# ip nhrp shortcut
R3#traceroute 192.168.1.2
R3#traceroute 192.168.1.2

Routing on Phase 3:

- o For EIGRP, need to disable split-horizon on Hub mGRE tunnel interface.
- o Disable Split-horizon on Hub tunnel Interface to ensure optimal forwarding.
- o Also need to disable the default EIGRP behavior to overwrite the next-hop as itself.
- o Issue summary-address command to force default route to be advertised out to spokes.

R1 Hub Router EIGRP Configuration

```
R1(config)#interface Tunnel 0
R1(config-if)# no ip split-horizon eigrp 1
R1(config-if)# no ip next-hop-self eigrp 1
R1(config-if)# ip summary-address eigrp 1 0.0.0.0 0.0.0.0
```

EIGRP Configuration on Hub Router

```
R1(config)#router eigrp 1
R1(config-router)#network 10.10.1.0 0.0.0.255
R1(config-router)#network 192.168.1.0
R1(config-router)#no auto-summary
```

EIGRP Configuration on Spoke Router

```
R2(config)#router eigrp 1
R2(config-router)#network 10.10.2.0 0.0.0.255
R2(config-router)#network 192.168.1.0
R2(config-router)#no auto-summary
```

EIGRP Configuration on Spoke Router

```
R3(config)#router eigrp 1
R3(config-router)#network 10.10.3.0 0.0.0.255
R3(config-router)#network 192.168.1.0
R3(config-router)#no auto-summary
```

```
R1# show ip route
```

```
R1# show ip route eigrp
```

```
R2# show ip route
```

```
R2# show ip route eigrp
```

```
R3# show ip route
```

```
R3# show ip route eigrp
```

```
R3# ping 10.10.2.1 source loopback 1
```

```
R2# ping 10.10.3.1 source loopback 1
```

```
R3# traceroute 10.10.2.1 source loopback 1
```

```
R3# traceroute 10.10.3.1 source loopback 1
```